25901443

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEPHY204 - MATERIAL CHARACTERISATION TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overalloutcome attainability.

Part A

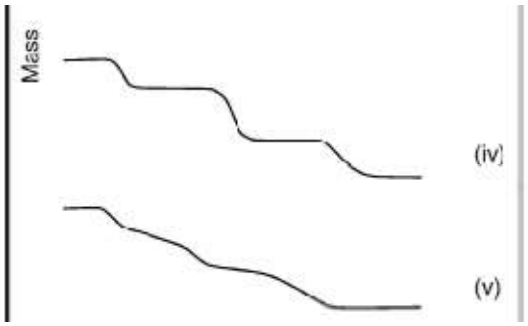
Short Answer

Answer any **10** questions

Each question carries **2** marks

1.	Write Bragg's law and explain the terms.		[K] / [CO1]
Ans.	$n\lambda = 2d \sin\theta$ n – order of reflection λ – wavelength of X-ray d – interplanar spacing θ – Bragg angle	1 mark for equation and 1 mark for explaining the terms.	
2.	Name any two commonly used laser sources in Raman spectroscopy.		[U] / [CO2]
Ans.	solid-state laser (ruby laser and Nd-YAG laser) gas laser (He–Ne laser and argon-ion laser) liquid laser (SeOCL2 laser and Europium Chelate laser) dye laser (Rhodamine 6G laser and Coumarin dye laser) semiconductor laser (GaAs laser and GaAsP laser).	2 marks for naming any two of the five laser sources listed.	
3.	Explain how conjugation affects the position of absorption maxima in UV–Visible spectrum		[U] / [CO2]
Ans.	Increasing the number of conjugated double bonds shifts the absorption maxima to longer wavelengths (e.g., 1, 5-hexadiene at 178 nm vs. 2,4-hexadiene at 227 nm)	give marks accordingly for any reasonable attempt. Examples not mandatory.	
4.	How are liquid samples prepared for infrared (IR) spectroscopy?		[U] / [CO2]
Ans.	A drop of a liquid organic compound is placed between a pair of polished sodium chlorideor potassium bromide plates, referred to as **salt plates.** When the plates are squeezed gently, a thinliquid film forms between them. A spectrum determined by this method	2 marks for the given answer, OR give marks accordingly for	

	is referred to as a **neat**spectrum since no solvent is used. Salt plates break easily and are water soluble. Organic compoundsanalyzed by this technique must be free of water. The pair of plates is inserted into a holderthat fits into the spectrometer.	any reasonable attempt. Examples not mandatory.	
5.	Why is the lifetime of phosphorescence greater than fluorescence?		[K] / [CO2]
Ans.	phosphorescence involves a forbidden "spin-flip" transition from an excited triplet state back to a singlet ground state **Beyond the scope of current syllabus.**	Give credits to any relevant answer on phosphorescence or fluorescence	
6.	What are the applications of Scanning Electron Microscopy (SEM) in materials science?		[U] / [CO3]
Ans.	SEM is used for microstructural and surface morphology examination.	2 marks for the given answer, OR give marks accordingly for any reasonable attempt.	
7.	Name the source used for generating electrons in an electron microscope.		[U] / [CO3]
Ans.	heated tungsten OR electron gun	2 marks for the given answer	
8.	Why do electron microscopes achieve higher resolution than opticalmicroscopes?		[U] / [CO3]
Ans.	The high resolution of electron microscopes results from the short wavelengths of the electrons used for microscope illumination. The wavelength of electrons in electron microscopes is about 10,000 times shorter than that of visible light.	2 marks for the given answer, OR give marks accordingly for any reasonable attempt.	
9.	What information is obtained in the spectroscopic mode of Scanning Tunneling Microscopy?		[U] / [CO3]
Ans.	surface properties such as superconduction and molecular adsorption on metal	2 marks for the given answer, OR give marks accordingly for any reasonable attempt.	
10	Explain phonons in the context of sample - electron beam interaction.		[U] / [CO3]
Ans.	Phonons are the quanta of atomic vibrations present in the case of a solid. A primary electron loses energy by exciting a phonon and causing the solid to heat up slightly. The electrons remain in the solid and gradually excite the phonons and hence heat the solid. The phonon scattering causes a large deflection of the primary electrons and hence can be used to study the material properties.	2 marks for the given answer, OR give marks accordingly for any reasonable attempt.	
11	What is the role of a temperature sensor in thermal analysis technique?		[U] / [CO4]

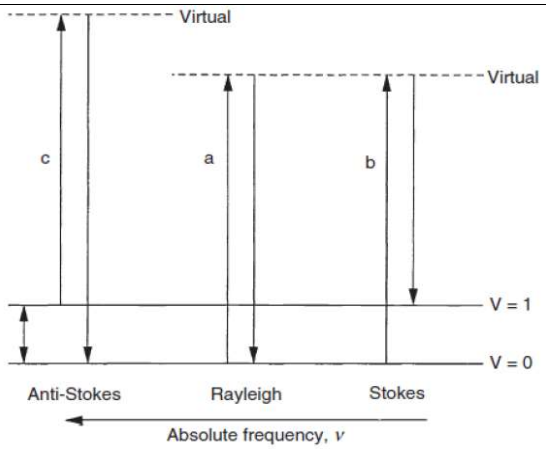
Ans.	The temperature sensor (thermocouple) measures the sample and/or reference temperature during heating or cooling. In TA instrumentation, the thermocouple is located near the sample holder but is rarely in direct contact with the sample itself — it is typically in contact with the sample holder. This means the measured temperature may not exactly represent the true sample temperature, especially at fast heating rates. The thermocouple output is used by the programmer and computer system to control the furnace and record the thermal response (Y-axis signal) as a function of temperature or time. Any thermal lag between the heating source and sample, caused by fast heating rates, leads to inaccurate temperature readings — making the sensor critical for reliable TA data.	2 marks for the given answer, OR give marks accordingly for any reasonable attempt.	
11	Sketch the thermogravimetric curve for multistage decomposition with and without a stable intermediate.		[U] / [CO4]
Ans.	• **With stable intermediate** —iv • **Without stable intermediate** (v) 	1 mark each	
12	Distinguish between Cahn microbalance and null point microbalance used in thermogravimetry.		[U] / [CO4]
Ans.	Both terms refer to the same instrument — the Cahn microbalance is a null-point type microbalance. The null-point microbalance maintains the sample in a fixed vertical position even as its mass changes, using a feedback system. The Cahn microbalance specifically senses vertical displacement of the sample via an optical system (light source, flag, light tube, and photodiode), and a feedback control system adjusts the current in a magnetic-coil system to keep the balance beam horizontal	2 marks for the given answer, OR give marks accordingly for any reasonable attempt.	
13	Give a short note on principle of Differential Thermal Analysis.		[U] / [CO4]
Ans.	DTA measures the temperature difference (ΔT) between a sample and an inert reference material as both are subjected to the same controlled heating or cooling program. The reference undergoes no thermal events, so any deviation in ΔT indicates a thermal event in the sample. The DTA curve is a plot of ΔT versus reference temperature or time. Endothermic events (e.g., melting) appear as downward peaks and exothermic events (e.g., crystallization) as upward peaks in a DTA curve.	2 marks for the given answer, OR give marks accordingly for any reasonable attempt.	

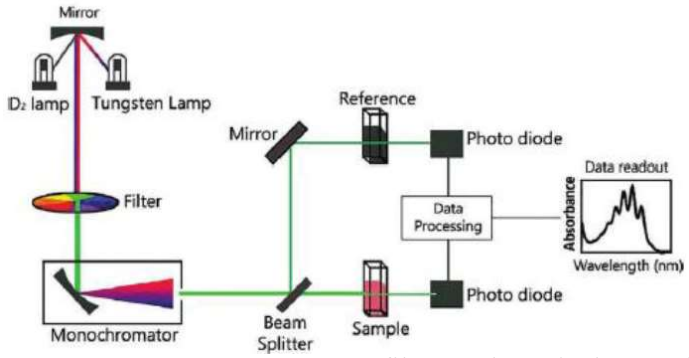
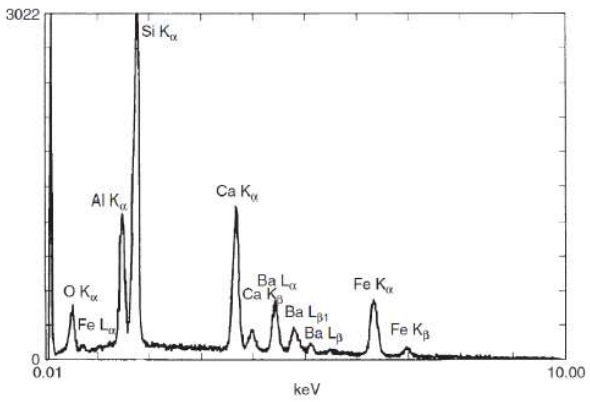
Part B

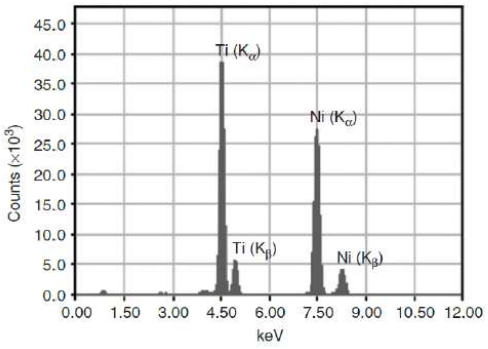
Short Essay

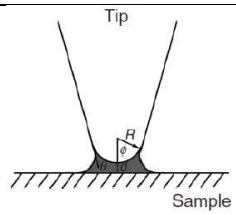
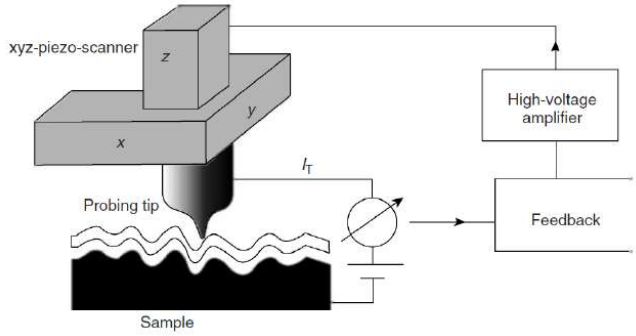
Answer any **6** questions

Each question carries **5** marks

14.	How can crystallite size be calculated from XRD peak broadening?		[U] / [CO1]
Ans.	The finite size of crystallites in a material causes broadening of X-ray diffraction peaks.Thebroadening is quantified by the Full Widthat Half Maximum (FWHM) of the diffractionpeak.The crystallite size is determined using the Scherrer equation: $D=K\lambda/\beta\cos\theta$ Where: D = crystallite size K = shape factor λ = wavelength of X-ray radiation β = FWHM θ = Bragg angle	5 marks for the given answer, OR give marks accordingly for any reasonable attempt.	
15.	Describe the mechanism of Raman scattering in detail.		[U] / [CO2]
Ans.	 A strong beam of laser from the ultraviolet–visible region (υo) is allowed to fall on the sample. The light that gets scatteredis of two categories. One is called Rayleigh scattering, which has the same frequency υo as that of the incident beam, while the other is called Raman scattering, which has a frequency that is either greater or less than the frequency of the incident beam. If the frequency of scattered light is less than the frequency of the incident beam ($\upsilon o-\upsilon m$, where υm is the frequency corresponding to the vibration of a molecule), then we have *Stokes lines*. If the frequency of the scattered light is greater than the frequency of the incident beam ($\upsilon o+\upsilon m$) then we have *anti-Stokes lines*.	5 marks for the given answer, OR give marks accordingly for any reasonable attempt.	
16.	Describe the main components of a UV spectrophotometer.		[K] / [CO2]

Ans.	 **Radiation Sources -** Tungsten filament lamp is the standard source, most effective between 320 nm and 1100 nm. Hydrogen discharge lamps provide a continuous spectrum from 185 nm to 380 nm. Deuterium arc lamps(190–370 nm) **Wavelength Selectors (Filters & Monochromators) -** Absorption filters use dyed glass or pigmented resins and have wide band widths (30–250 nm). Interference filters are more accurate, providing a narrow band pass of approximately 10 nm by using a dielectric spacer. Monochromators: These use either prisms (refractive or reflective types) or gratings to disperse light. **Sample Compartment and Cuvettes -** Optical Glass (335–2500 nm), Quartz (Far-UV) (170–2700 nm). **Detectors -** Barrier Layer Cell, Photo Emissive Cell, Photomultiplier Tube (PMT)	5 marks for the given answer, OR give marks accordingly for any reasonable attempt.	
17.	What insights can we gain from qualitative analysis of EDS data? Explain with a graph.		[U] / [CO3]
Ans.	From qualitative analysis we can identify elemental composition in the specimen. AND  Figure 6.10 EDS spectrum of glass which includes Si, O, Ca, Al, Fe, and Ba. (Repro- An EDS spectrum is presented as the intensity of characteristic X-ray lines across the X-ray energy range.Figure 6.10 shows the EDS spectrum of a glass specimen containing multiple elements including Si, O, Ca, Fe, Al, and Ba in an energy range up to 10 keV.	5 marks for the given answer, OR give marks accordingly for any reasonable attempt.	

	OR  Figure 6.17 EDS spectrum of a Ni–Ti alloy obtained in an XRF spectrometer. Figure 6.17 shows an XRF spectrum (EDS type) for a Ni–Ti alloy. The spectrum shows one major *Kα* peak and one minor *Kβ* peak for each of Ni and Ti. The *Kα* includes *Kα*1 and *Kα*2, which are often overlapped in the spectrum due to the small difference in their energy levels. The relative positions and intensities of the *α* and *β* lines also can help us to identify chemical elements if there is a possibility that the *Kα* or *Kβ* line of one element overlaps another element's *L* lines. We should note that the primary X-ray source of XRF includes both continuous and characteristic radiation		
18.	Summarize how capillary forces affect tip-sample interactions in an atomic force microscope.		[U] / [CO3]
Ans.	Capillary forces are forces resulting from water vapor condensation between tip and sample. In an air environment, water vapor is likely to form a condensed nucleus there. The water nucleus condensed between tip and sample will form a meniscus, as schematically shown in Figure 5.6, similar to the water surface in a capillary tube. The capillary force on the tip can be estimated $$F_{\text{cap}} = \frac{4\pi R\gamma \cos\theta}{1 + d/[R(1-\cos\phi)]}$$ γ is the surface tension of a water nucleus. The meanings of angles θ and φ, and d and R are indicated in figure. For a tip radius of 100 nm, the maximum capillary force reaches about 90 nN. This force is significantly larger than other forces being discussed. If an attractive force of the order of 10−8 to 10−7N is recorded, it most likely arises from the capillary. Capillary forces play an important role in AFM measurements in air. We can reduce these forces by covering the sample with hydrophobic molecules. AFM measurements in liquid are not affected by capillary forces.	5 marks for the given answer, OR give marks accordingly for any reasonable attempt.	

			
19.	Explain the functions of probe and scanner in Scanning Probe Microscopy.		[U] / [CO3]
Ans.	In an SPM, the probe is commonly made from tungsten wire. The tungsten wire conducts the tunneling current flowing between surface atoms and tip atoms.A probe motion sensor senses the spacing between the tip apex and the sample and provides a correction signal to control the piezoelectric scanner movement. In the SPM, the tunneling current is used to sense the spacing between the probe and the sample surface.The scanner controls the probe that moves over the sample in three dimensions in a precise way. The scanning probe needs to be positioned with an accuracy of 1pm (10−12 m) if atomic resolution is required. To achieve this level of precision, a scanner is made from piezoelectric materials. *Piezoelectric materials change shape in an electric field or generate an electric field due to a shape change. The relative length of the change is proportional to applied voltage (E).* $\frac{\Delta L}{L} = d_{ij}E$ *where dij is the piezoelectric coefficient; its typical value is about 0.3 nm mm−1 V−1. Thus, greater than 300 V is required to achieve 1μm movement for a 10-mm long piezoelectric tube.* 	5 marks for the given answer, OR give marks accordingly for any reasonable attempt.	
20.	Distinguish between conventional Differential Scanning Calorimetry andtemperature-modulated Differential Scanning Calorimetry.		[A] / [CO4]
Ans.	• In conventional DSC, the sample temperature changes linearly with time according to $T = T_0 + \beta t$, where T_0 is the starting temperature, t is time, and β is the constant heating rate. The heat flow at any moment is expressed as $dH/dt = C_p\beta + f(T,t)$, where $C_p\beta$ represents reversing heat flow and f(T,t) the nonreversing (kinetic) component. However, conventional DSC measures only the total heat flow and cannot separate these two components. • In TMDSC, a sinusoidal (cyclic) temperature modulation is superimposed on the constant heating rate: $T = T_0 + \beta t + B \sin(\omega t)$,	Explanation on conventional DSC 2.5 marks Temperature-Modulated DSC 2.5	

	where B is the amplitude of modulation (±1 to ±10 K) and $\omega = 2\pi/p$ (p = modulation period, typically 10–100 s). This creates both a constant heating profile (giving total heat flow) and a sinusoidal heating profile (giving information on the rate of temperature change). • The main advantage of TMDSC is its ability to separate reversing and nonreversing thermal events. Reversing events include glass transition and melting (fusion). Nonreversing events include oxidation, curing, relaxation, and cold crystallization. Conventional DSC cannot distinguish overlapping thermal events, whereas TMDSC clearly resolves them into separate curves. • Raw TMDSC data are more complex than conventional DSC data and require deconvolution to generate standard DSC curves. The reversing heat-flow curve and nonreversing heat-flow curve are extracted separately. The total curve from TMDSC is equivalent to the conventional DSC curve.	marks	
21.	Explain the solid phase transformation and thermal decomposition.Give example.		[A] / [CO4]
Ans.	• Solid phase transformation refers to a first-order transition in which a material changes its crystal structure without melting. Such transitions involve an enthalpy change and are therefore detectable by DTA and DSC as peaks in the thermal curve. The curve peaks of DTA and DSC readily reveal solid-state phase transformations over a temperature range. • For example pure iron changes its crystal structure from body-centered cubic (BCC) to face-centered cubic (FCC) when the temperature increases to 912 °C.. • Thermal decomposition is the chemical breakdown of a material upon heating, resulting in a mass change (usually mass loss due to release of volatile products). It is primarily studied using thermogravimetry (TG), where the sample mass is continuously monitored as a function of temperature. Decomposition is characterized by two temperatures: T_i (onset of mass change) and Tf (completion of mass change). • For example, solid $CaCO_3$ will decompose to form solid CaO and gaseous CO_2 over a certain temperature range. • Both phenomena can be studied simultaneously using combined TG-DSC/DTA techniques. While TG monitors mass change (decomposition), DSC/DTA monitors heat flow (both phase transformation and decomposition). For example, in polymer composite analysis, TG reveals decomposition temperatures and ceramic content (as in HA/UHMWPE composites, Figure 10.32), while DSC can reveal any solid-state phase transitions occurring within the same temperature range. Together, they provide a comprehensive thermal characterization of a material.	Solid phase transformation with any example 2.5 mark thermal decomposition with any example 2.5 mark	